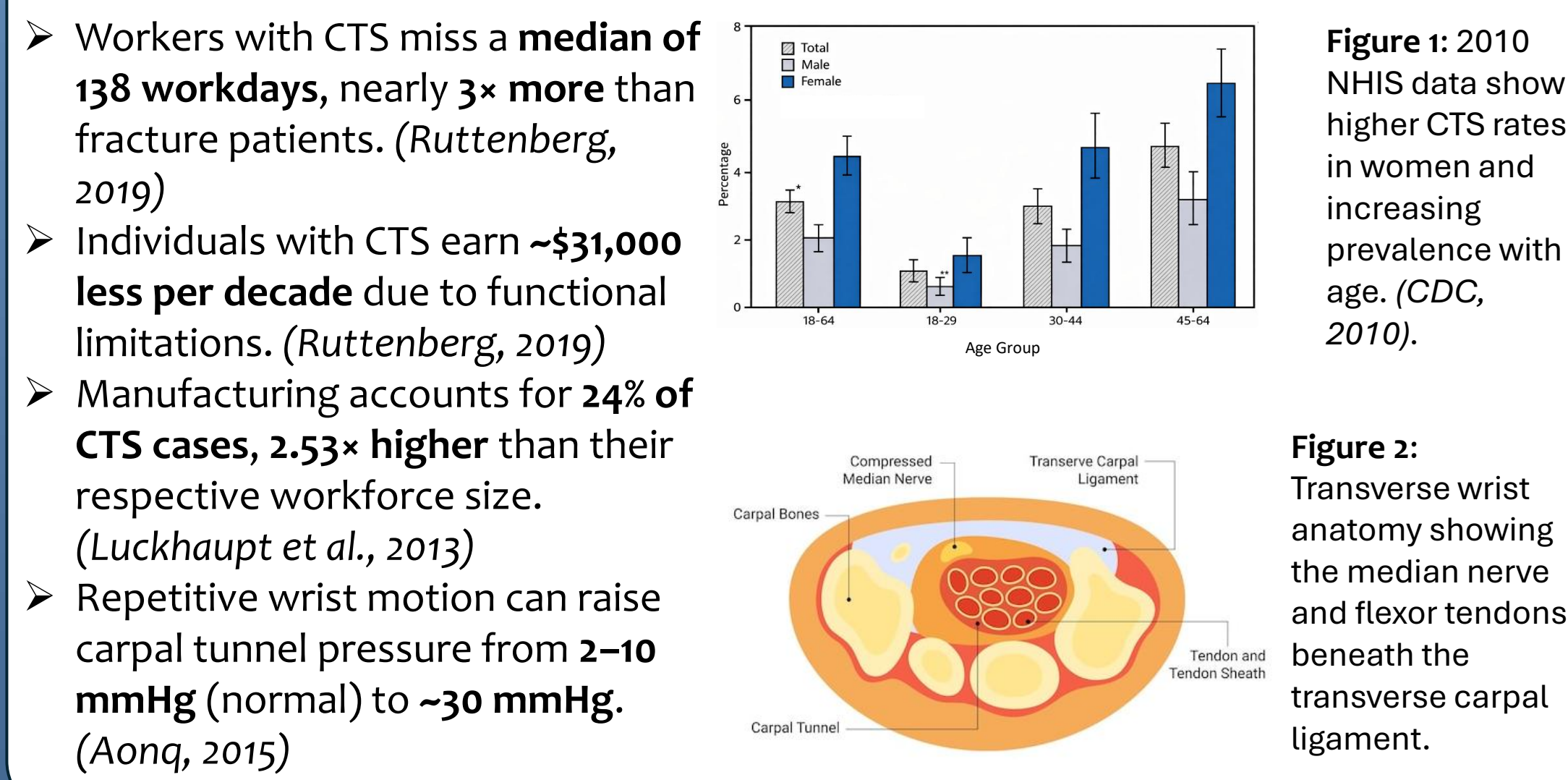
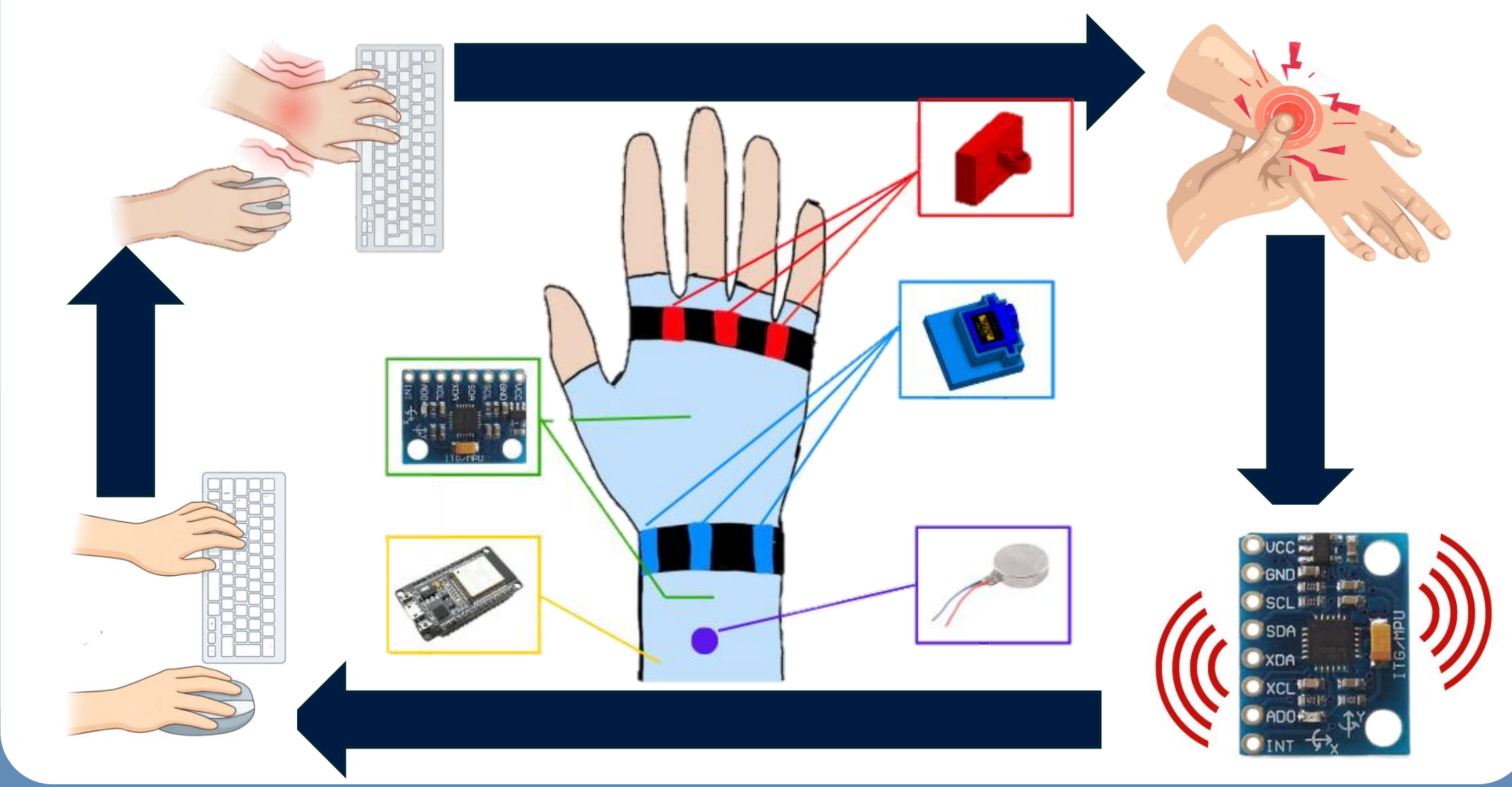
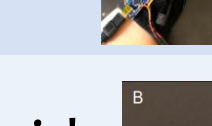
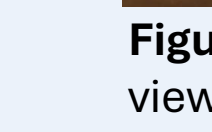


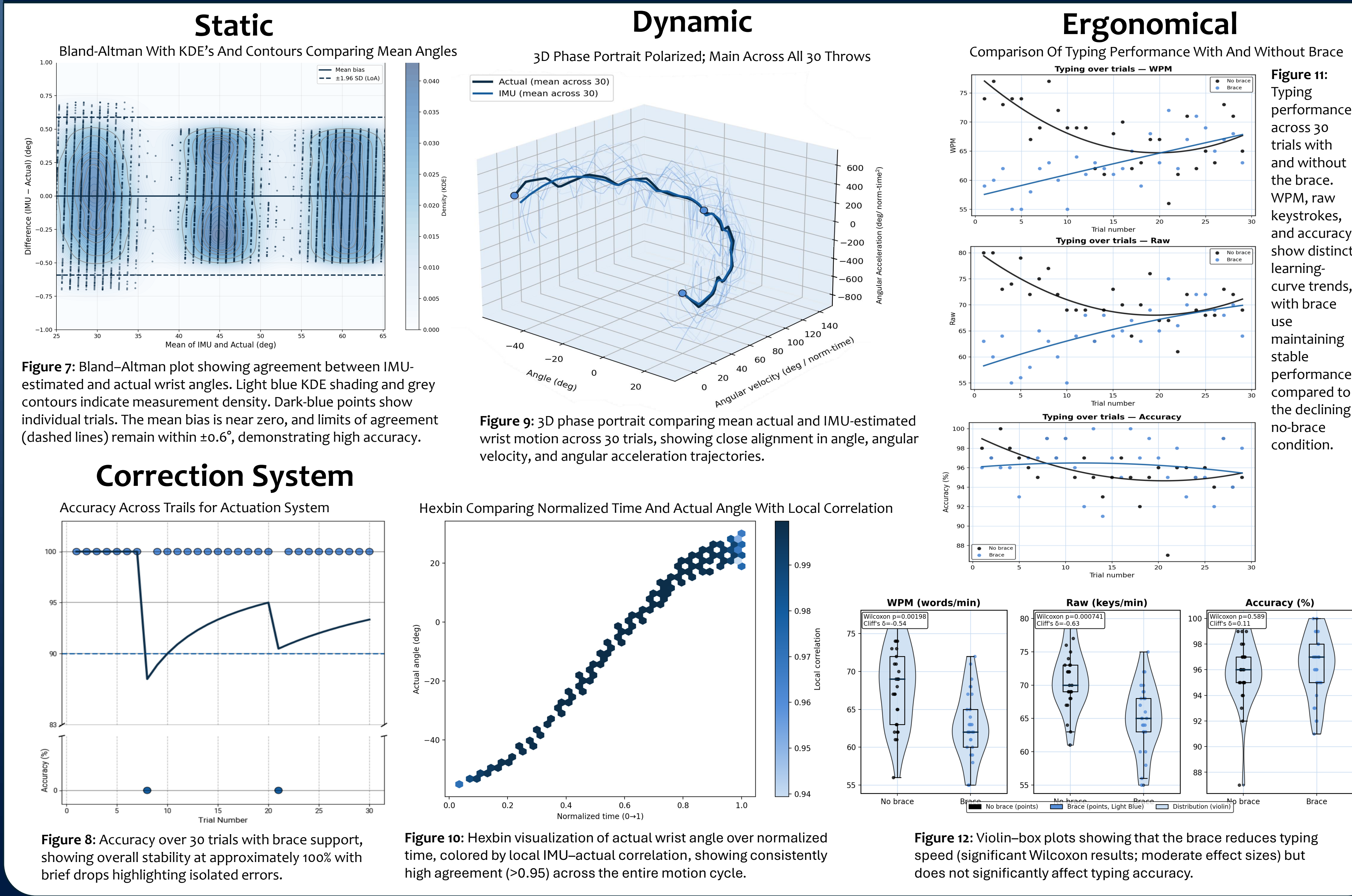
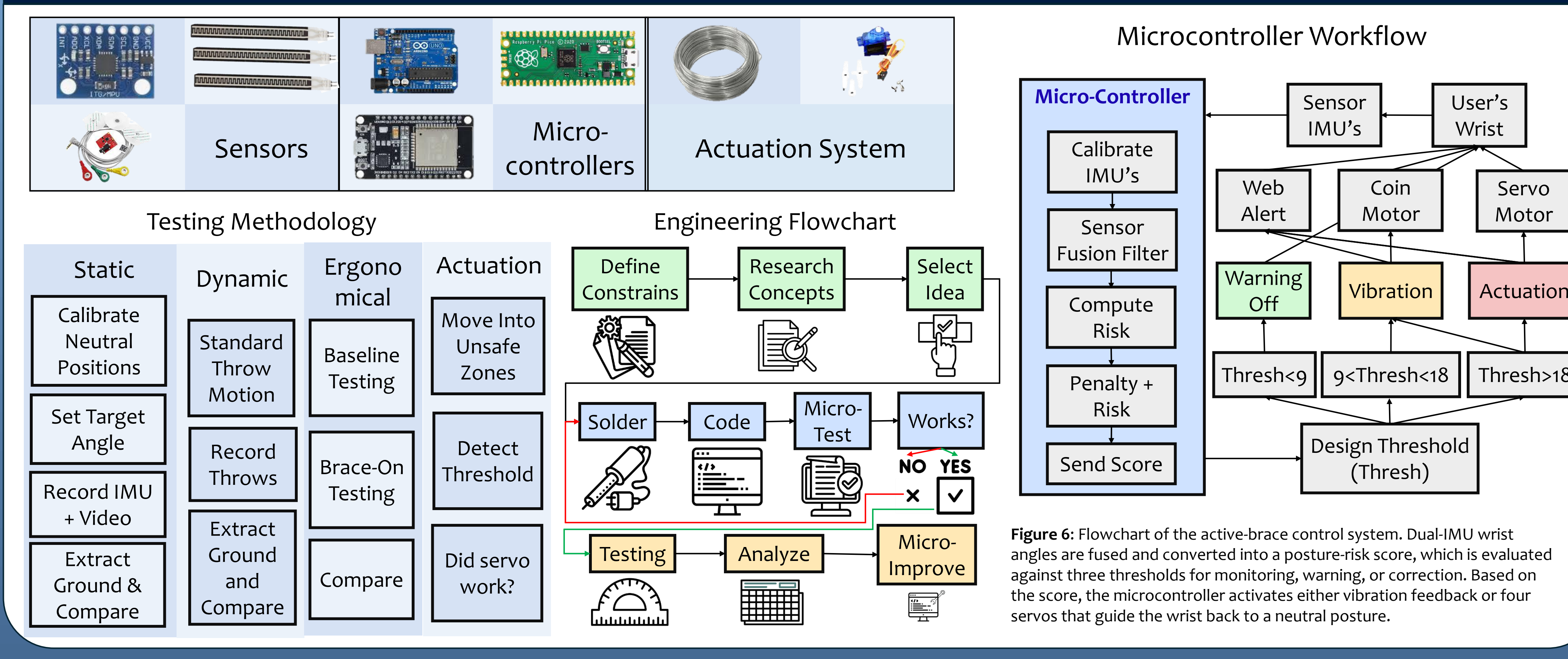


Prior Work	How it Works	Strengths	Limitations/Gaps	Photo of Brace
Aryapet yan et al. (2021)	Two IMUs + EMG calculate overuse score and trigger alerts	➤ Integrates IMU+EMG ➤ Real-time warnings	➤ Understand threshold EMG signal instability	 Figure 3: Microcontroller with dual IMUs, EMG sensor, vibration motor, and FSR
Piper (2017)	Rigid rotary-encoder brace validated with VICON	➤ High angular accuracy ➤ Strong Validation	➤ Bulky and rigid ➤ Unsuitable for continuous daily wear	 Figure 4: Lateral views showing side profile of brace
De Angelis et al, (2019)	Passive soft brace limits wrist posture and reduces pressure	➤ Clinically shown to reduce symptom and pressure	➤ Passive ➤ Does not promote long term habit formation.	 Figure 5: Dorsal views showing the strap, prism pad, and dorsal band

Criteria	Idea 1	Idea 2	Idea 3	Passive Splint	Active Research
Posture Detection Accuracy (25)	20	20	20	0	22
CTS Risk Reduction Effectiveness (25)	18	18	18	23	16
Real-Time Feedback Responsiveness (15)	10	14	14	6	13
Active Correction Capability (10)	0	8	8	4	5
User Comfort/Wearability (10)	7	4	9	10	9
System Reliability & Consistency (10)	3	7	7	10	8
Cost & Accessibility (10)	10	4	6	8	6
Expandability & Future Integration (5)	4	4	4	2	5
Total (110):	72	79	86	63	84
Topsis (1):	0.61	0.71	0.76	0.33	0.73

There is no wearable device that can accurately measure wrist angle in real time, detect unsafe wrist positions, and provide immediate feedback.

Create a wrist brace that accurately tracks wrist angles in real time, identifies unsafe positions, and provides alerts to guide the wrist back to a safer posture.



Statistical Test (Flex/Ext) (Rad/Ulnar) (Throwing) (Typing WPM)				
DESCRIPTIVE STATISTICS				
Mean Absolute Error	0.253°	0.263°	0.391°	5.17 WPM (down 7.6%)
Standard Deviation	0.292°	0.311°	0.501°	±4.44 WPM
HYPOTHESIS TESTING				
t-tests	5/6 trials: p > 0.05 1/6: p=0.005*	All 6 trials: p > 0.10	t(978) = 5.045 p = 4.59×10 ^{-7***}	t(29) = 3.367 p = 0.002**
One-way ANOVA (Across angle targets)	F(2,5997)=1.626 p = 0.197	F(2,5959)=2.437 p = 0.087	—	—
EFFECT SIZE ANALYSIS				
Cohen's d	d = -0.020 (negligible)	—	d = 0.135 (small)	d = 1.06 (large)
Pearson Correlation (r)	r = 0.9997***	r = 0.9997***	r = 0.9998***	—
R² (Variance Explained)	99.94%	99.95%	99.96%	—
VARIANCE & DISTRIBUTION ANALYSIS				
Shapiro-Wilk Test (Normality)	All p < 0.05 Platykurtic (k ≈ -1.2)	All p < 0.05 Platykurtic	—	p > 0.05 (Normal)
CONFIDENCE INTERVALS & ACCURACY				
95% Confidence Interval	r: [0.9996, 0.9998]	r: [0.9996, 0.9998]	—	WPM: [2.16, 8.18] ↓
Accuracy within ±0.5°	100.00%	93.34%	73.08%	—
Accuracy within ±1.0°	100.00%	100.00%	94.33%	—

Conclusion	Future Work
➤ Exceptional accuracy: $r > 0.999$ correlation with clinical goniometer; 100% of static measurements within $\pm 0.5^\circ$ ➤ Dynamic validation: 94.33% accuracy maintained during throwing motion despite 2.73× higher variance ➤ Consistent performance: No significant variation across 60-30° range of motion (ANOVA $p > 0.05$) ➤ Acceptable trade-offs: 7.6% typing speed reduction with preserved accuracy; 93.33% actuation reliability ➤ Clinical readiness: All measurements within $\pm 1.0^\circ$ clinical standard; validated for therapeutic applications	➤ Soft Actuation: safer, compliant wrist correction ➤ EMG Fusion: muscle + motion risk scoring ➤ Micro Exoskeleton: guided wrist posture assistance ➤ Motion Validation: high-precision mo-cap comparison ➤ Wearability Upgrade: flexible, comfortable sensor housing ➤ Risk Prediction: forecasting CTS overuse patterns ➤ Adaptive Feedback: smarter vibration and force cues ➤ Ergo Dashboard: personalized posture analytics display

Arayatheg, N., Miao, W., & Bourkka, K. (2011). Over-use-injury model for carpal tunnel syndrome (augmenting human expertise). University of California, Berkeley.

<https://escholarship.org/uc/item/9590721>

Centers for Disease Control and Prevention. (2019). *Prevalence of self-reported carpal tunnel syndrome among employed adults aged 18-64 years who had carpal tunnel syndrome in the past 12 months, by sex and age group* – National Health Interview Survey, 2010. *Morbidity and Mortality Weekly Report*, 68(40), 1702. <https://www.cdc.gov/mmwr/preview/mmwrhtml/mm6840a3.html>

De Angelis, C., Pierdelfino, F., & Luchetti, L. (2008). Prevalence of a self-reported carpal tunnel syndrome and the association with carpal tunnel syndrome: A generalized carpal tunnel syndrome. *Acta Neurologica Scandinavica*, 119(1), 67–72. <https://doi.org/10.1111/j.1469-7580.2008.01072.x>

Luchetti, S., E., Dahlbaker, J. M., Ward, B. W., Sweeney, M. H., Settling, J. P., & Calver, D. M. (2019). Prevalence and work-relatedness of carpal tunnel syndrome in the working population of the United States. *Journal of Occupational Medicine*, 56(6), 615–619. <https://doi.org/10.1093/oxford/oi.kz009>

Kahn, K., & McIn, C. (1989). Design of a wearable device to monitor wrist posture to prevent carpal tunnel syndrome. *Journal of Medical Engineering and Technology*, 4(3), 389–398. <https://doi.org/10.1080/03090902.1989.11638267>

Manfredi, M., & Cazzulani, A. (2019). Carpal tunnel syndrome. *Work*, 63(1), 1–10. <https://doi.org/10.1080/13632539.2019.1638267>

McDonald, J., Stanescu, T., Drachov, C., & MacCalli, L. (2013). Changes in the carpal tunnel while wearing the Manus® soft hand brace: A sonographic study. *Journal of Hand Surgery (European Volume)*, 38(1), 67–70. <https://doi.org/10.1177/1753122213501246>

Patten, R. J. (2017). Design and validation of an instrumented rigid wrist orthosis for high-precision motion tracking. *Biomedical Engineering Online*, 16(1), 112. <https://doi.org/10.1186/s12938-017-0015-7>

Ritter, R. (2019). Design and social and economic impact of carpal tunnel syndrome among maintenance-of-way employees. *Journal of Ergonomics*, 9(1), Article 246. <https://doi.org/10.24267/2474-3628.246>

Umar, R. Z., Ahmed, R., Rahman, M. A., & Zuhri, N. (2019). Development and validation of an IMU-based posture assessment system for ergonomic risk evaluation. *Applied Ergonomics*, 80, 123–132. <https://doi.org/10.1016/j.apergo.2019.05.010>